

PHY

PHY Interfacing — Summary

What is PHY?

The **Physical Memory Interface (PHY)** is the hardware layer that sits between the digital Memory Controller (MC) and the external DRAM chips. Think of it as a translator — it converts the logical memory commands from the controller into actual electrical signals on the memory bus. The connection between the MC and PHY is standardized through the **DFI (DDR PHY Interface)**.

Architecture Overview

The DDR PHY is broken into four main sections:

1. Byte Slices

Each byte slice handles 8 bits of data and contains:

- **DQS (Data Strobe)** — acts as a local clock for data, contains Gate Training, Write Leveling, and DLL circuits
- **DQ (Data lines)** — the actual data bits, with Write/Read Data Eye centering per rank and Jitter Measurement

2. Master Timing Functions

The "brain" section that handles system-wide timing and training:

- **PLL** — generates the core clock
- **Clock Deskew** — corrects timing skew across signals
- **VREF Training** — finds the optimal voltage threshold
- **Eye Diagram TDR** — signal integrity analysis
- **Loop Back / PRBS** — self-test capability
- Also contains FIFOs and training interfaces for the MC

3. Memory Controller Interface

The FIFOs that buffer communication between the MC and PHY:

- Write FIFO, Read FIFO, Command Address FIFO, Training Interface

4. Command/Address Block + Customized I/O Ring

- **Command/Address Block** — uses a Multi-Cycle DLL and supports SDR/DDR tuning for sending commands to DRAM
 - **Customized I/O Ring** — the outermost analog layer; contains drivers and receivers that physically push signals onto PCB traces toward RAM. Handles **Impedance Matching (ZQ)** to prevent signal reflections.
-

PHY Control Block (PUB/PCTL)

This is the conductor of the entire PHY — it runs the training sequences at boot:

Training Procedures

1. **Write Leveling** — Compensates for PCB trace length differences. The PHY sends test pulses and adjusts DLL delay taps until the DQS strobe aligns with the DRAM clock (Feedback = 1 → LOCKED).
2. **DQS Gate Training** — Since DQS floats during idle, the PHY needs to know exactly when to "open the gate" to capture incoming data. Opening too early captures noise; too late misses data. Training finds the precise window.
3. **VREF Training** — Finds the optimal voltage reference (V_{th}) that maximizes the data eye opening. The controller sends test patterns and the system sets VREF to the center of the passing voltage range, maximizing noise margin.

Data Eye Diagram

A graphical tool for analyzing **signal integrity**. The open space in the center (the "eye") is the safe zone where data can be reliably sampled. A **mask test** checks that signals don't intrude into the diamond-shaped mask in the center.

- **Open eye** = no mask violation = good signal
- **Closed eye** = mask violation = signal integrity problem

Read Data Eye Centering (DQ-DQS Centering)

After gate training, the PHY further refines the DQS strobe position to sit exactly at the center of the valid DQ data window:

- DQS too far left → FAIL
- DQS too far right → FAIL
- DQS centered → PASS (optimal)

DDR PHY Interface (DFI)

DFI is the **standardized interface** between the Memory Controller and the PHY — a universal language so components from different vendors can interoperate without

redesigning the interface each time.

DFI Signal Groups

Group	Signals
Control	address, bank, CAS, CKE, CS, ODT, RAS, WE, reset
Write Data	wrdata, wrdata_en, wrdata_mask
Read Data	rddata_en → rddata, rddata_dnv, rddata_valid
Update	ctrlupd_req/ack, phyupd_req/ack
Status	init_start, init_complete, freq_ratio, parity
Training	rdlvl and wrlvl signals (read/write leveling)
Low Power	lp_req, lp_wakeup, lp_ack

DFI 5.0/5.1 — Key Upgrades

Three major pillars introduced:

1. **PHY Independent Boot Sequence** — In older versions the controller micromanaged boot. Now the controller just sends a single "Start" command and the PHY handles initialization itself.
2. **Expanded Frequency Range Support** — Supports Frequency Set Points (FSPs) so the system can hop between low and high speeds dynamically (e.g., for battery saving in mobile).
3. **New PHY-to-Controller Interaction** — Bidirectional packet-based communication synced to a fast Write Clock (WCK). The PHY can now instantly alert the controller about errors like **Command Address Parity** errors.

PHY Configuration Registers

Timing

- **tRAS** (Row Address Strobe delay), **CL** (CAS Latency), **CWL** (CAS Write Latency), **tWTR** (Write-to-Read delay)

Delay / Training

- Per-bit DQ delays, DQS read/write delays, gate training window, VREF values, CA training offset

Impedance & Drive Strength

- ODT (On-Die Termination) enable/resistance, drive strength, slew rate control

Mode Register Settings (MRW/MRR)

- Burst Length (BL8/BC4), RTT termination resistance, data mask enable, bank grouping

Refresh Control

DRAM cells leak charge and must be periodically refreshed. Four modes:

Mode	Description
Auto Refresh	Refresh every tREFI (typically 7.8 μs), takes tREFC time
Self Refresh	DRAM refreshes itself while controller clock is off — used in low power states
Per Bank Refresh	One bank at a time based on need — reduces performance penalty
Fine Granularity (FGR)	Smaller, more frequent refreshes instead of one big chunk every 7.8 μs

Mode	tREFI	tREFC
1x	7.8 μs	350 ns
2x	3.9 μs	260 ns
4x	1.95 μs	160–200 ns

Key Takeaway: The PHY is a sophisticated analog/digital bridge. Most of its complexity lies in the **training sequences** that run at boot to compensate for real-world physical imperfections like PCB trace delays, signal reflections, and voltage noise — ensuring reliable high-speed data transfer to DRAM.